The Wisdom of Crowds

MSIS 521

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Ox?

- Francis Galton, 1906

Weight? 1,198 pounds

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Challenger

- January 28, 1986
- Stock market reactions:
  - Rockwell International -6% → -3%
  - Lockhead -5% → -3%
  - Martin Marietta -3% → -3%
  - Morton Thiokol -6% → -12%
- Truth revealed 6 months later
  - O-ring seal
- The market was smart!

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Madness of the Crowds

- In 19th century, it was
  - "The madness of the crowds" (Mackay, 1841)
  - "A sensible individual, as a member of a crowd becomes a blockhead" (Baruch)
  - "Madness is the exception in individuals and the rule in groups" (Nietzsche)
  - "Crowds deliver verdicts of which every individual disagree" (Le Bon)

Were they wrong?

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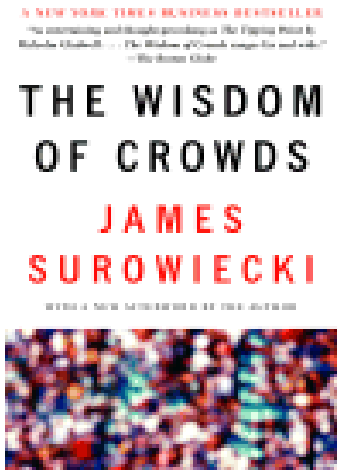
Wisdom vs. Stupidity of the Crowds

- To get "wisdom" from a crowd we need:
  - **Diversity of opinion** (same inputs, each participant weighing differently)
  - **Independence** (no influencing on each other's decisions)
  - **Decentralization** (different inputs + bottom-up process)
  - **Aggregation** mechanism
    - Reputation/Reward/Quality control
    - Encourage micro-contributions
- If some of the above fail, we get stupidity:
  - No independence: bubbles
  - No diversity: no appreciation of some inputs
  - No decentralization: no access to vital information
  - No aggregation: the blind men and the elephant



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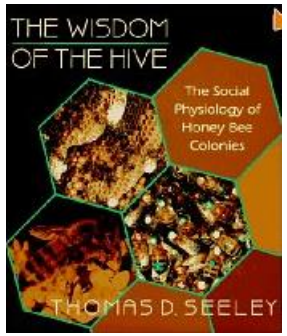
Diversity

- A hive of bees
- Efficient at finding food
- Scouts to many directions



- Smartness required?
  - Experiments by Scott Page
  - Mix of smart and not-so-smart

- Chase the expert
  - “Who the hell wants to hear actors talk?”
  - “I think there is a world market for maybe five computers.”

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Independence

- Information cascades / herding
  - William Beebe's observation in Guyana forests
  - "circular mill"
    - Army ants
    - Simple rule: follow the ant in front
    - Circle of 1200 feet in circumference
  - Are we ants?

- Milgram, Bickman, and Berkowitz experiment
  - 1, 5, 15

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Decentralization

- Individual decision making
- Tacit knowledge
- Strength
  - Independence, specialization, coordination
- Weakness
  - No guarantee on information transfer

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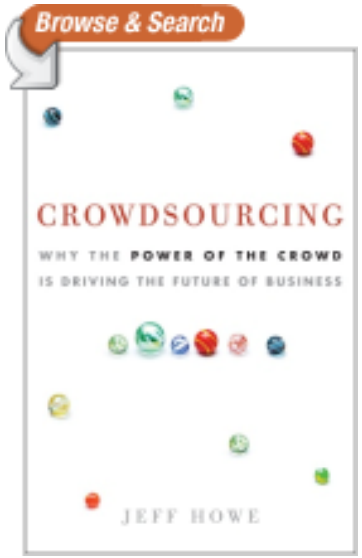
Coordination

- "No one goes there anymore. It is too crowded."
  - Yogi Berra said of Toots Shor's nightclub
- Do we need a central planner?
- Brian Arthur experiment
  - El Farol problem
  - Have fun if the bar is < 60% full.
  - Agents' strategies
  - 100 weeks, the bar was, on average, exactly 60% full

- Thomas Schelling experiment with law students (focal point)

Crowdsourcing

- Definition
  - Crowdsourcing is the act of taking a job traditionally performed by a designated agent (usually an employee) and outsourcing it to an undefined, generally large group of people in the form of an open call.
- Types
  - Creation
  - Invention
  - Organization
  - Prediction



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Wisdom of Crowds in Modern AI

Many core ideas of “wisdom of crowds” appear in modern AI and data science:

- Ensemble methods (e.g., random forests, XGBoost)
  - Random forests, gradient boosting, and other ensembles combine many weak models.
  - Diverse, partially independent predictors + aggregation often beat a single complex model.
- Crowd labels and human-in-the-loop learning
  - LLMs are fine-tuned on feedback from many human annotators, not one expert.
  - Aggregation of ratings helps shape model behavior.

Takeaway: designing AI systems often means designing good “crowds” of models and people, and good ways to aggregate them.

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Threadless

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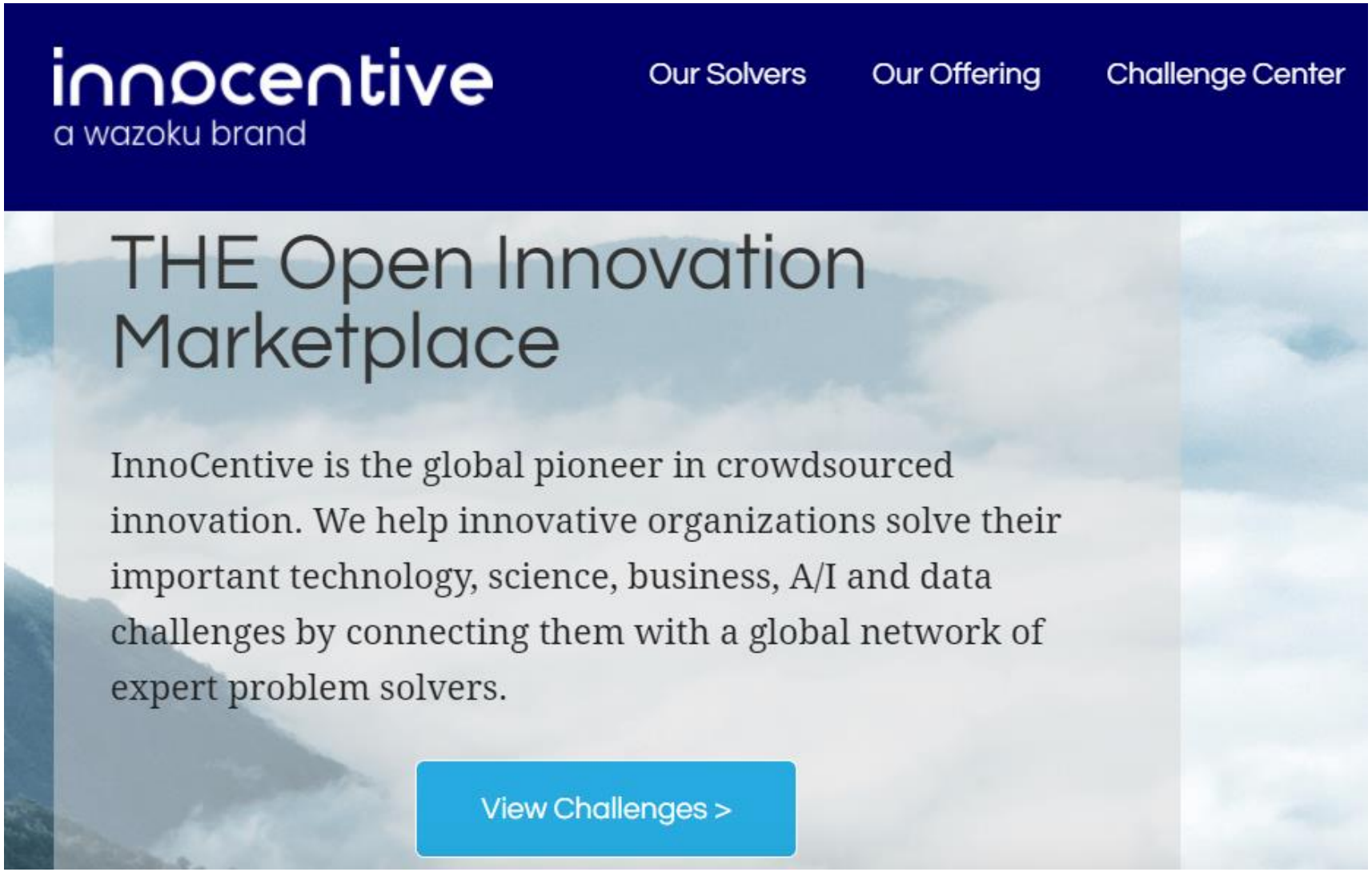
Case Study: InnoCentive



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InnoCentive

- Started in 2001 by a VP of Eli Lilly

- A Craigslist for solutions to scientific problems
  - Posters post problem and bounty
  - Solvers try to find a solution
  - Double blind process, identities hidden

- More than 1/3 of problems solved
  - 57% “reduction to practice”
  - Problems unsolvable by labs of P&G, etc.

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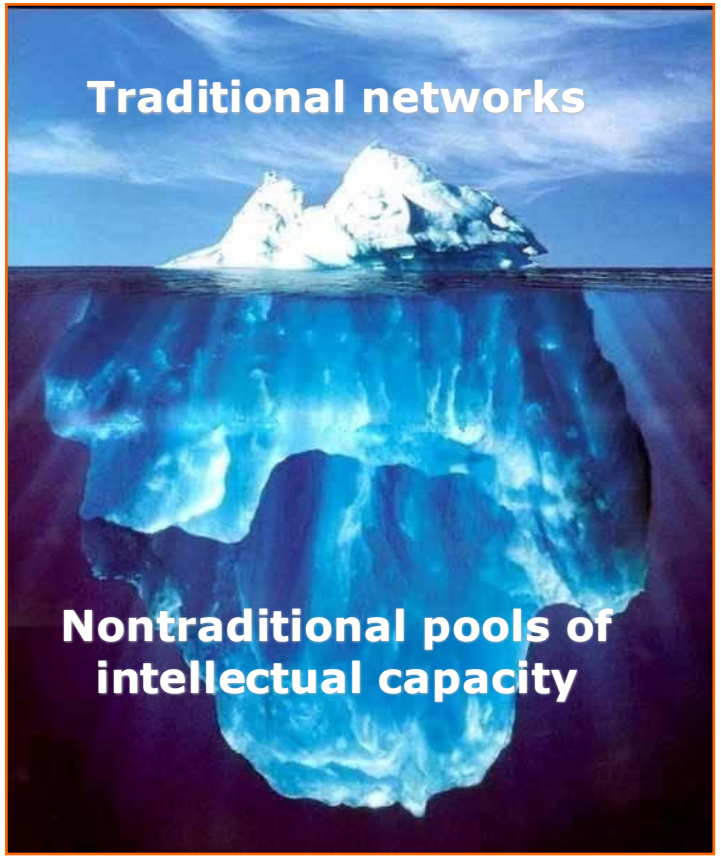
Innocentive Reaches Talent Pool

"Wherever you work, most of the smart people are somewhere else." --- Bill Joy

- Traditional pools
  - US, EU Academics
  - Contract labs (FTE)
  - Individual networks

- Opportunity pools
  - **Global Academia**
  - **Researchers in Russia, India, China.**
  - **Scientists in other industries**
  - **Excess capacity**
  - **Retirees**



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Solver Community

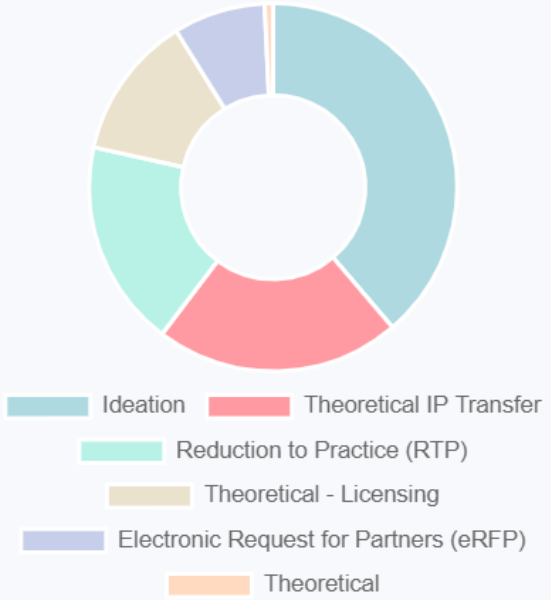
Top Solvers - Most Challenges Won 2021

Ra...	Name (show more)	Challenges Won
1	Juan Martin	2
2	Anonymous	1
2	SATYA PRAKASH TRIPATHI	1
2	Anonymous	1
2	Jayesh Vani	1

Top Solvers - Most Submissions 2021

Ra...	Name (show more)	Submissions
1	Anonymous	24
2	JATINDER HANSPAL	10
3	Anonymous	9
3	Yaroslav Granowski	9
5	Martin Grigorov	7

Challenges by Type



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Challenges

Excavating and Repairing Low Voltage Underground Networks

Electronic Request for Partners...

Open until 18 May 2021

Awa... Collaboration with En...

VIEW CHALLENGE

Habitat for Humanity Challenge: Malaria Prevention through Innovations i...

Reduction to Practice (RTP)

Open until 07 Jun 2021

Award $20,000

Active Solvers 82

VIEW CHALLENGE

Market Sentiment Analysis

Reduction to Practice (RTP)

Open until 03 May 2021

Award $20,000

Active Solvers 118

VIEW CHALLENGE

Cleaner Indoor Air During Wildfires

Theoretical

Open until 17 May 2021

Award $50,000

Active Solvers 233

VIEW CHALLENGE

A2A Challenge: Fast Intelligence from Flowing News

Reduction to Practice (RTP)

Open until 13 Apr 2021

Award $40,000

Active Solvers 131

VIEW CHALLENGE

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InnoCentive's Value Proposition

- Seekers
  - Project Roadblock
  - Revive Stalled Projects or Close "Dead" Projects
  - Culture – work outside the walls

- Solvers
  - Intellectual challenge
  - Business development opportunities
  - Financial reward
  - Recognition

We see similar models on Kaggle, Foldit, Zooniverse, open-source projects, etc., where small contributions from many people solve problems that competitive labs struggle with.

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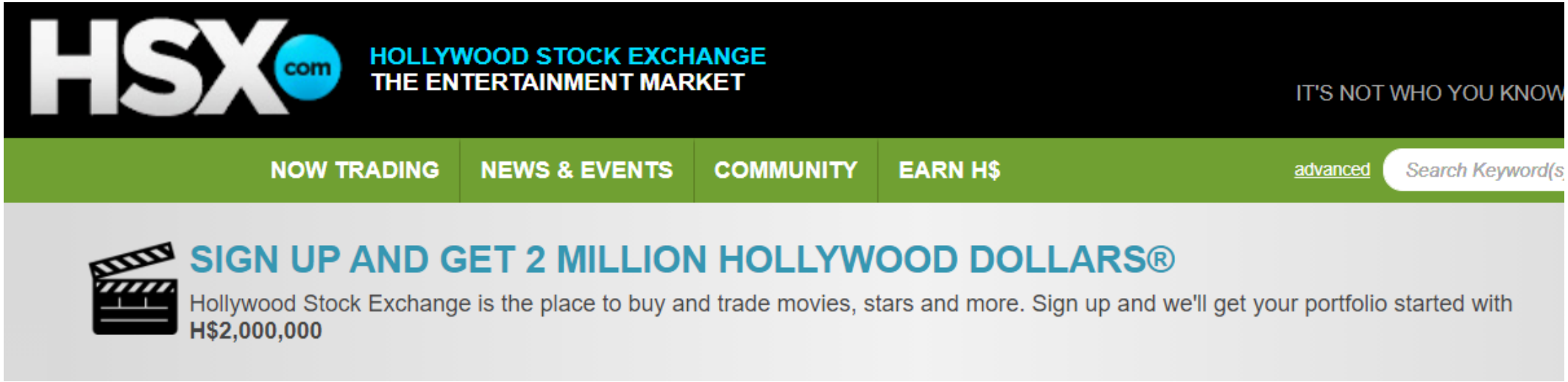
Prediction Markets

- A place where information is aggregated via market (or other) mechanisms for the primary purpose of forecasting events, or the probability that an event will occur



Some other examples of prediction markets where people spend and earn actual money:
Kalshi and Novig (mainly sports-related prediction markets)
Polymarket (prediction market for a variety of real-world events)

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Practical Problems

- Achieving **independence**:
  - Independence assumes no interaction
  - Independence assumes all have information at same time
  - People imitate “early winners” (esp. after 2 observations)
  - **Catch**: Doing “what the group” does is often good!
- Achieving **diversity** and **decentralization**:
  - Difficult when the group aims to satisfy individuals
  - When personalized results are preferred, the (single result) of homogenous groups work better
- **Aggregation**
  - Difficult to solicit actions from crowd
  - Attention budget limited
  - Models that capture implicit actions inherently more scalable

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Non-Market Alternatives vs. Markets

- Opinion polls
  - Sampling
  - No incentive to be truthful
  - Equally weighted information
  - Hard to be real-time
- Ask Experts
  - Identifying experts can be hard
  - Incentives
  - Combining opinions can be difficult
- Machine learning/Statistics
  - Historical data
  - Past and future are related
  - Hard to incorporate recent new information

- Prediction Markets
  - Self-selection
  - Monetary incentive and more
  - Money-weighted information
  - Real-time
  - Self-organizing

- Prediction Markets
  - No need for data
  - No assumption on past and future
  - Immediately incorporate new information

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